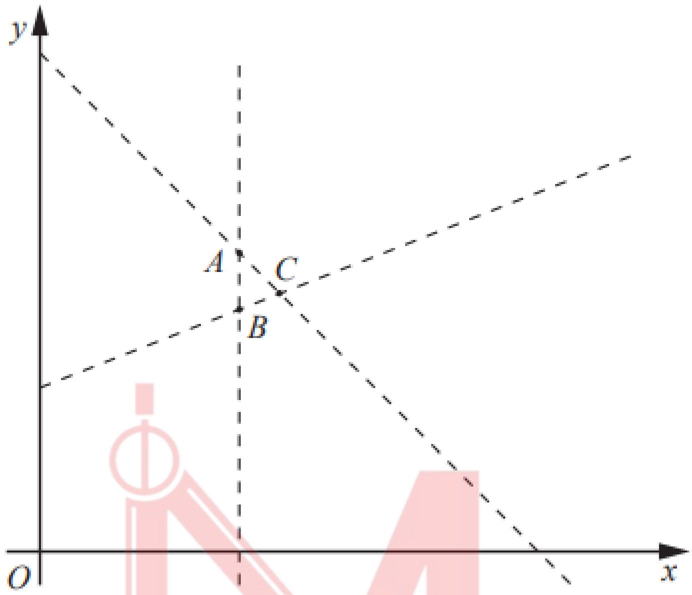
Worksheet 2

1. **W14/qp12/q23**



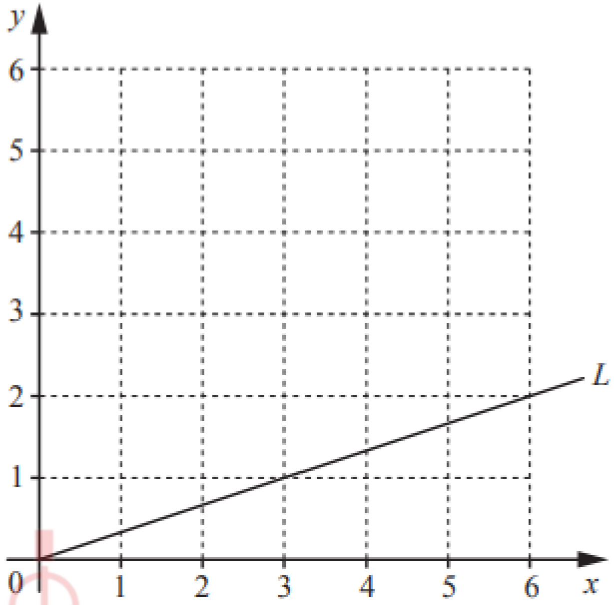
The diagram shows the three lines $x = 8$, $x + y = 21$ and $2y = 12 + x$ which intersect at the points A, B and C.

a. Find the coordinates of B. [1]

b. The region inside triangle ABC is defined by three inequalities. One of these is $x + y < 21$. Write down the other two inequalities. [2]

c. Find the coordinates of the point, with integer coordinates, that is inside triangle ABC. [1]

2. S15/qp11/q21



a. On the grid, shade the region, R, given by these inequalities. [2]

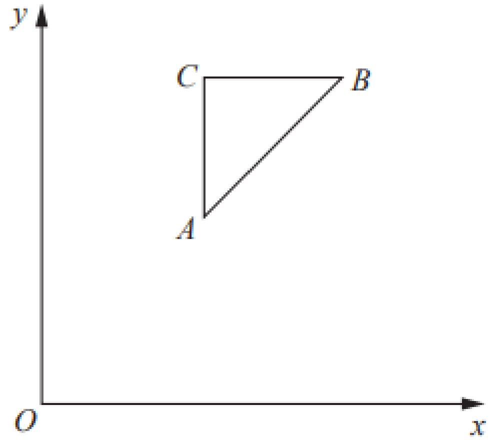
b. The line L, with equation $y = \frac{1}{3}x$, is drawn on the grid.

c.

i. Draw the line $y = \frac{1}{3}x + k$ so that it passes through a point belonging to R such that k is as large as possible. [1]

ii. Write down this largest value of k. [1]

3. W15/qp12/q18

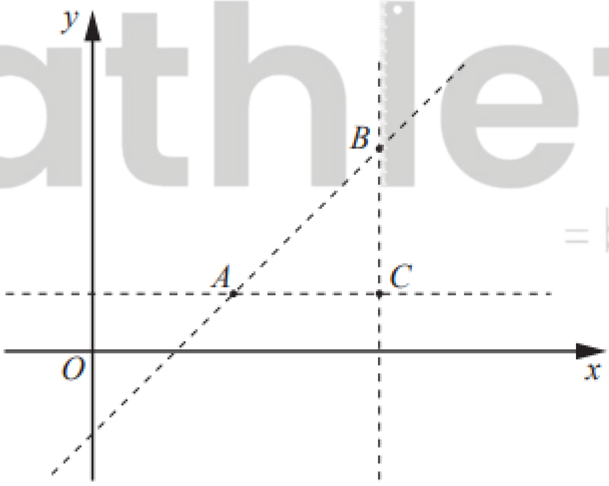


The sides of the triangle ABC are formed by the straight lines with equations $x = 3, y = 6, y = x + \frac{1}{2}$.

a. The region inside the triangle is defined by three inequalities. Write down these three inequalities. [2]

b. The point (4, k), where k is an integer, lies inside the triangle. Find the value of k. [1]

4. W16/qp11/q24



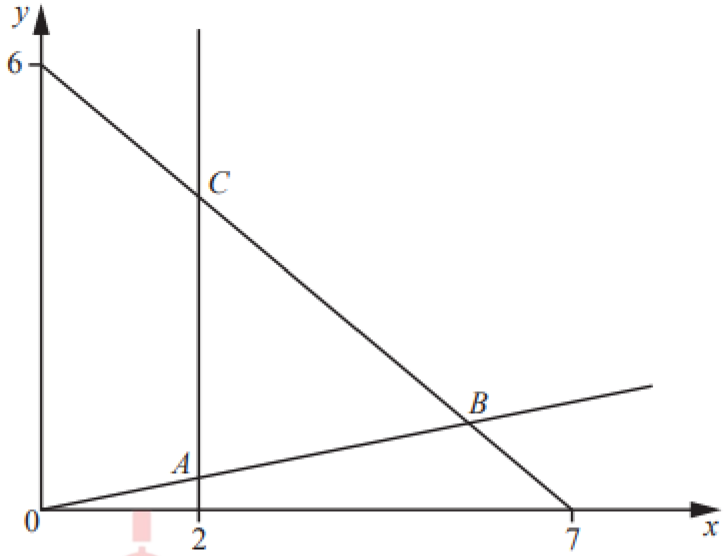
A is the point (5, 2) and B is the point (9, 6). AC is parallel to the x-axis. CB is parallel to the y-axis. The equation of the line AB is $x - y = 3$.

a. Find the coordinates of C. [1]

b. The region inside triangle ABC is defined by three inequalities. Write down these inequalities. [3]

c. The point (a, b), where a and b are integers, lies inside triangle ABC. It also lies on the line $y = \frac{1}{2}x$. Find the value of a and the value of b. [2]

5. W18/qp11/q21



In the diagram, the equation of the line

- through B and C is $6x + 7y = 42$
- through A and B is $y = \frac{x}{5}$.

a. The region inside triangle ABC is defined by three inequalities. One of these is $y > \frac{x}{5}$. Write down the other two inequalities. [2]

b. The line $y = kx$ passes through triangle ABC. Find all the possible integer values of k. [2]

Answer Key:

1. (a)(8, 10) (b)$x > 8 \; oe \; 2y > 12 + x$ (c) (9, 11)

2. (a) Correct region identified (b) (i) Line parallel to L, through top left hand point of R (ii) 3.5 to 4 (inclusive)

3. (a) $x > 3$; $y < 6$; $y > x + 1/2$; oe (b) 5

4. (a) (9, 2) (b) $x < 9 \; oe \; y > 2 \; oe \; x - y > 3 \; oe$ (c) a = 8 b = 4

5. (a) $x > 2 \; oe \; and \; 6x + 7y < 42$ (b) Both 1 and 2, only, nfww